Understanding Router Interrogation Results

Investigator Reference Guide

This section explains what the collected information means and why it matters.

Default Gateway

What It Is

The default gateway is usually the router.

Examples:

192.168.1.1
192.168.0.1
10.0.0.1

Why It Matters

The default gateway is the device that directs traffic between the local network and the Internet.

It is usually the primary device being investigated.

Local Subnet

What It Is

The range of addresses used by devices on the network.

Examples:

192.168.1.0/24

192.168.0.0/24

Why It Matters

The subnet determines which devices belong to the local network.

All discovered devices should generally fall within this range.

DNS Server

What It Is

A server that translates names into IP addresses.

Example:

google.com

becomes

142.250.x.x

Why It Matters

DNS activity may reveal services that were accessed.

Examples:

- Google
- Snapchat
- Discord
- iCloud
- Find My iPhone

ARP Cache

What It Is

A list of devices recently communicating on the network.

Why It Matters

Provides:

- IP address
- MAC address

May identify:

- Phones
- Computers
- Cameras
- Printers

ARP entries often reveal devices before investigators log into the router.

MAC Address

What It Is

A unique hardware identifier assigned to a network device.

Example:

F4:5C:89:12:34:56

Why It Matters

MAC addresses are often more reliable than IP addresses.

IP addresses change.

MAC addresses usually remain consistent.

Useful for identifying:

- Phones
- Laptops
- Cameras
- Smart TVs

Hostname

What It Is

The device name.

Examples:

Kris-iPhone
Samsung-S24
Titus-iPad
HP-Laptop
RingCam01

Why It Matters

Hostnames often associate a device with a person.

One of the quickest ways to identify ownership.

Nmap Host Discovery

What It Is

A list of devices currently responding on the network.

Why It Matters

Provides:

- IP address
- MAC address
- Sometimes device manufacturer

Useful for determining:

- What devices are present right now
- Unknown devices

- Hidden network equipment

Open Ports

What It Is

Network services exposed by a device.

Examples:

80
443
22
23

Why It Matters

Open ports tell investigators what services are running.

Examples:

Port	Common Service
80	Web Interface
443	Secure Web Interface
22	SSH
23	Telnet

53	DNS
3389	Remote Desktop
5900	VNC

Router Uptime

What It Is

The amount of time since the router last rebooted.

Why It Matters

May indicate:

- Power outage
- Manual reboot
- Factory reset
- Firmware update

A reboot shortly before an incident may be important.

DHCP Lease

What It Is

A record showing when a device received an IP address.

Why It Matters

One of the most valuable router artifacts.

Can help answer:

- Was a device present?
- When did it arrive?
- When did it leave?

DHCP History

What It Is

Historical lease assignments.

Why It Matters

Shows devices that may no longer be connected.

Often useful for:

- Missing person cases
- Homicides
- ICAC investigations

Wireless Client History

What It Is

A record of devices connecting to WiFi.

Why It Matters

May show:

- Connection time
- Disconnect time
- Last seen time

Often more valuable than current device lists.

Last Seen

What It Is

The most recent time a device communicated with the router.

Why It Matters

Frequently becomes part of a timeline.

Example:

Victim Phone
Last Seen:
05/31/2026 14:22

DNS Log

What It Is

A record of domain name lookups.

Why It Matters

May identify services used before an event.

Examples:

icloud.com
findmy.apple.com
maps.google.com

Useful for developing leads.

Administrative Log

What It Is

A record of router administration activity.

Why It Matters

May reveal:

- Logins
- Failed logins
- Password changes
- Configuration changes

Useful for determining whether someone altered the router.

Firewall Log

What It Is

A record of allowed or blocked traffic.

Why It Matters

May reveal:

- Remote access attempts
- External connections
- Scanning activity

Port Forwarding

What It Is

A rule allowing Internet access to devices inside the network.

Why It Matters

May expose:

- Cameras
- DVRs
- Computers
- NAS devices

Useful when investigating remote access.

UPnP

What It Is

A feature that automatically creates port forwarding rules.

Why It Matters

Can expose devices to the Internet without the owner's knowledge.

VPN

What It Is

A Virtual Private Network.

Examples:

- WireGuard
- OpenVPN
- IPSec
- Tailscale

Why It Matters

May affect attribution and routing.

Its presence alone is not evidence of criminal activity.

WAN IP Address

What It Is

The public IP assigned by the ISP.

Why It Matters

Can be used for:

- ISP records
- Subscriber information
- Search warrants
- Timeline development

Document it immediately.

NAS Device

What It Is

Network Attached Storage.

Examples:

- Synology
- QNAP
- WD My Cloud

Why It Matters

Often contains years of files, backups, photographs, and videos.

May be more valuable than the router itself.

NVR / DVR

What It Is

Video recording systems for surveillance cameras.

Why It Matters

May contain:

- Video evidence
- Motion events
- Timestamps
- User activity

Frequently identified through router interrogation.

Packet Capture

What It Is

A recording of network traffic.

Why It Matters

Captures:

- DNS
- DHCP
- ARP
- Device communications

Useful for understanding live network activity.

Router Configuration Backup

What It Is

An export of router settings.

Why It Matters

Preserves:

- Wireless settings
- DHCP settings
- Port forwarding
- VPN settings
- Firewall rules

Often the most complete snapshot of the router’s configuration.